Yifan Zeng

Ph.D. Student · RL for LLM Agents

zengyif@oregonstate.edu xhmy.github.io Google Scholar LinkedIn

Education

Oregon State University Corvallis, OR
Ph.D. in Computer Science 2023–2029 *(expected)*
- Advisor: Prof. Huazheng Wang
- Research Areas: Robust LLM Reasoning & Decision Making, Agentic AI

Zhuhai College of Jilin University Zhuhai, China
B.E. in Data Science 2019–2023
- GPA: 3.84/4.00, Rank: 1/249

Industry Experience

Research Scientist Intern | Adobe Jun 2026 – Sep 2026
- Designed an optimization method for agent skill libraries that refines the partition and granularity of each skill, improving skill- selection accuracy and token efficiency of the Adobe Experience Platform Agent Orchestrator and other Adobe agent harnesses

AI Agent Research Intern | AG2AI Jun 2025 – Sep 2025
- Completed the SimpleDoc paper: designed a multi-agent RAG pipeline for long-document understanding, covering reasoning-based retrieval, ranking, and question answering

Applied Scientist | Quanbao Network Technology Apr 2022 – Aug 2023
- Medical imaging: applied image pretraining and object detection to anatomical landmark detection in bronchoscopy images, and deployed the model on endoscopic devices for real-time monitoring that helps physicians track the bronchoscope's position
- Rail defect detection: proposed a self-supervised pretraining method for railway ultrasonic inspection data and built the object detection training and inference framework on labeled defect data, achieving significantly higher accuracy than existing commercial solutions

Academic Experience

Oregon State University Mar 2024 – Present
Graduate Research Assistant Mar 2024 – Present
- Built RLHF pipelines (SFT, reward model training, preference optimization with TRL) and distributed online RL training pipelines (GRPO/verl) for LLM alignment and LLM agent workflows
- Studied robustness of LLM-based ranking, retrieval, and decision-making systems

Graduate Teaching Assistant Sep 2024 – Jun 2025, Sep 2025 – Present
- Supported instruction for graduate and undergraduate computer science courses, including grading and student support
- TA for the AI Capstone course, mentoring graduate students in developing AI-based products, including LLM-agent RAG systems for funding and patent search

Skills

RL & Alignment: Multi-Agent RL for LLM Workflows, Distributed Online RL (GRPO, Multi-Node DP/TP Training), RLHF (SFT, Reward Modeling, Preference Optimization), Inference-Time Alignment (Best-of-N, Tree Search); built with verl, TRL, HF Transformers, vLLM

LLM Agents & Retrieval: Multi-Agent Systems, Agent Optimization, Agent Skill Libraries, Multi-Modal RAG, LLM-based Ranking, Web-Agent Trajectory Evaluation

Safety & Robustness: Jailbreak Defense, Prompt-Injection Attacks on LLM Rankers, Ranking Consistency

Efficient ML & Vision: Sparse Fine-Tuning & Inference, Self-Supervised Pretraining, Object Detection; built with PyTorch

Publications

1. Jiale Liu, Huajun Xi, Shaokun Zhang, **Yifan Zeng**, Tianwei Yue, Chi Wang, Jian Kang, Qingyun Wu, Huazheng Wang. **Who&When Pro: Can LLMs Really Attribute Failures in AI Agents?**, NeurIPS 2026.
2. **Yifan Zeng**, Yiran Wu, Yaolun Zhang, Wentian Zhao, Kun Wan, Qingyun Wu, Huazheng Wang. **When Does Multi-Agent RL Improve LLM Workflows? Workflow, Scale, and Policy-Sharing Tradeoffs**, *arXiv preprint arXiv:2605.24202*, 2026.
3. Haoyu Zheng, **Yifan Zeng**, Linghao Song, Murali Emani, Wenqian Dong. **Enabling Unstructured Sparse Fine-Tuning and Inference for Foundation Models on Wafer-Scale Engine**, SC '25 Workshops.
4. Yaoyao Qian, Yuanli Wang, Jinda Zhang, Yun Zong, Meixu Chen, Hanhan Zhou, Jindan Huang, **Yifan Zeng**, Xinyu Hu, Chan Hee Song, Danqing Zhang. **WebGraphEval: Multi-Turn Trajectory Evaluation for Web Agents using Graph Representation**, NeurIPS 2025 Workshop MTI-LLM.
5. Chelsi Jain*, Yiran Wu*, **Yifan Zeng***, Jiale Liu, Shengyu Dai, Zhenwen Shao, Qingyun Wu, Huazheng Wang. **SimpleDoc: Multi-Modal Document Understanding with Dual-Cue Page Retrieval and Iterative Refinement**, EMNLP 2025 (Main Conference).
6. Jiale Liu, **Yifan Zeng**, Shaokun Zhang, Chi Zhang, Malte Højmark-Bertelsen, Marie Normann Gadeberg, Huazheng Wang, Qingyun Wu. **Divide, Optimize, Merge: Fine-Grained LLM Agent Optimization at Scale**, EMNLP 2025 (Findings).
7. Jiahao Qiu*, Yifu Lu*, **Yifan Zeng**, Jiacheng Guo, Jiayi Geng, Chenhao Zhu, Xinzhe Juan, Ling Yang, Huazheng Wang, Kaixuan Huang, Yue Wu, Mengdi Wang. **TreeBoN: Enhancing Inference-Time Alignment with Speculative Tree-Search and Best-of-N Sampling**, EMNLP 2025 (Findings).
8. Hui Yuan*, **Yifan Zeng***, Yue Wu*, Huazheng Wang, Mengdi Wang, Liu Leqi*. **A Common Pitfall of Margin-based Language Model Alignment: Gradient Entanglement**, ICLR 2025.
9. Yaoyao Qian*, **Yifan Zeng***, Yuchao Jiang, Chelsi Jain, Huazheng Wang. **The Ranking Blind Spot: Decision Hijacking in LLM-based Text Ranking**, EMNLP 2025 (Main Conference).
10. **Yifan Zeng***, Ojas Tendolkar*, Raymond Baartmans*, Qingyun Wu, Lizhong Chen, Huazheng Wang. **LLM-RankFusion: Mitigating Intrinsic Inconsistency in LLM-based Ranking**, Transactions on Machine Learning Research (TMLR), 2024.
11. **Yifan Zeng***, Yiran Wu*, Xiao Zhang, Huazheng Wang, Qingyun Wu. **AutoDefense: Multi-Agent LLM Defense against Jailbreak Attacks**, NeurIPS 2024 Workshop SafeGenAI.
12. Jianqi Yan*, Junjie Luo*, **Yifan Zeng**, Alex P Leung, Jie Feng, Hong-Hao Zhang, Weipeng Lin. **Modeling the Time Evolution of Compact Binary Systems with Machine Learning**, The Astrophysical Journal, 973 (2), 163, 2024.

* denotes equal contribution.